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### **FILTER MODULE FOR IMAGING APPARATUS**

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#### **Abstract**

A filter module usable in conjunction with an imaging apparatus including an imaging element. The filter module includes a first optical filter; a first support member supporting the first optical filter; and a first drive unit moving the first support member. The filter module is configured to be positioned along an optical axis of the imaging element included in the imaging apparatus, the first optical filter being rotationally movable by the first drive unit between a first position and a second position. The first position is a position within an imaging range of the imaging element, and the second position is a position out of the imaging range of the imaging element.

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## **Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation under 35 USC § 120 of co-pending U.S. application Ser. No. 19/065,345, filed Feb. 27, 2025, which is a continuation of U.S. application Ser. No. 17/994,046, filed Nov. 25, 2022, now U.S. Pat. No. 12,289,512 B2, which claims priority under 35 USC § 119 to Japan Application No. 2022-001089, filed Jan. 6, 2022, the entire disclosures of which are each incorporated herein by reference.

### **BACKGROUND OF THE INVENTION**

#### **Field of the Invention**

[0002] The present disclosure relates to an imaging apparatus, and more particularly to a filter module for an imaging apparatus.

#### **Description of the Related Art**

[0003] For example, WO 2019/155908 A discloses an imaging apparatus including a neutral density (ND) filter unit whose light transmittance can be changed. The ND filter unit includes a plurality of ND filters having different light transmittances, and a rotary disk that supports the plurality of ND filters. The plurality of ND filters are provided in the rotary disk so as to be arranged in the circumferential direction about the rotation center line of the rotary disk. As the rotary disk rotates, one ND filter is disposed in front of an imaging element.

### **SUMMARY OF THE INVENTION**

[0004] However, in the case of the imaging apparatus described in WO 2019/155908 A, a retraction space for a plurality of ND filters that are not used is required, and as a result, the imaging apparatus becomes large.

[0005] Therefore, an object of the present disclosure is to enable an optical filter having various light transmittances to be used in an imaging apparatus without increasing the size of an imaging apparatus.

[0006] In order to solve the above problem, according to one aspect of the present disclosure, there is provided an imaging apparatus including: an imaging element including an imaging surface on which light from a subject is incident; a first filter unit including an electronic first optical filter capable of changing a light transmittance; and a first drive mechanism that rotates the first filter unit about a first rotation center line between a first filtering position and a first retraction position, in which the first filtering position is a position where the first optical filter exists in front of the imaging surface of the imaging element and the light before reaching the imaging surface passes through the first optical filter, and in which the first retracted position is a position where the first filter unit is out of the front of the imaging surface.

[0007] According to the present disclosure, an optical filter having various light transmittances can be used in an imaging apparatus without increasing the size of the imaging apparatus.

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## **Description**

## BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic front perspective view of an imaging apparatus according to an embodiment of the present disclosure;

[0009] FIG. 2 is a rear perspective view of a filter module;

[0010] FIG. 3 is an exploded rear perspective view of the filter module;

[0011] FIG. 4A is a rear view of the filter module in a state where a first filter unit is located at a first filtering position and a second filter unit is located at a second retraction position;

[0012] FIG. 4B is a rear view of the filter module in a state where the first filter unit is located at a first retraction position and the second filter unit is located at a second filtering position;

[0013] FIG. 5A is a front view schematically illustrating a layout of the imaging apparatus of the embodiment;

[0014] FIG. 5B is a front view schematically illustrating a layout of an imaging apparatus of Comparative Example 1;

[0015] FIG. 5C is a front view schematically illustrating a layout of an imaging apparatus of Comparative Example 2;

[0016] FIG. 6A is a rear view of the filter module in a state where the second filter unit rotates toward the second filtering position while the first filter unit rotates toward the first retraction position;

[0017] FIG. 6B is a rear view of the filter module illustrating a rotation state of the first and second filter units subsequent to FIG. 6A;

[0018] FIG. 6C is a rear view of the filter module illustrating a rotation state of the first and second filter units subsequent to FIG. 6B;

[0019] FIG. 6D is a rear view of the filter module illustrating a rotation state of the first and second filter units subsequent to FIG. 6C;

[0020] FIG. 7 is an exploded rear perspective view of the first filter unit;

[0021] FIG. 8 is a front perspective view of a ND filter assembly; and

[0022] FIG. 9 is a partial cross-sectional view of the ND filter assembly.

## DESCRIPTION OF THE PREFERRED EMBODIMENT

[0023] Hereinafter, an embodiment will be described in detail with reference to the drawings as appropriate. However, unnecessarily detailed description may be omitted. For example, detailed description of already well-known matters and redundant description of substantially the same configuration may be omitted. This is to avoid the following description from becoming unnecessary redundant and to facilitate understanding by those skilled in the art. In addition, the inventor(s) provides the accompanying drawings and the following description to enable those skilled in the art to sufficiently understand the present disclosure, which does not intend to limit the claimed subject matter.

[0024] Hereinafter, an imaging apparatus according to an embodiment of the present disclosure will be described with reference to the drawings.

[0025] FIG. 1 is a schematic front perspective view of an imaging apparatus according to an embodiment of the present disclosure. Note that the X-Y-Z orthogonal coordinate system illustrated in the drawings is for facilitating understanding of the embodiment of the present disclosure, and does not limit the embodiment of the present disclosure. The X-axis direction is a front-rear direction of the imaging apparatus, the Y-axis direction is a left-right direction, and the Z-axis direction is a height direction. Note that a side where a subject is present at the time of capturing is defined as a front side of the imaging apparatus.

[0026] As illustrated in FIG. 1, an imaging apparatus **10** according to an embodiment of the present disclosure includes a filter module **12**. The imaging apparatus **10** is a so-called single-lens reflex camera.

[0027] FIG. 2 is a rear perspective view of the filter module. FIG. 3 is an exploded rear perspective

view of the filter module.

[0028] As illustrated in FIGS. 2 and 3, in the case of the present embodiment, the filter module **12** includes a casing **14**, a first filter unit **16**, and a second filter unit **18**.

[0029] The casing **14** is made of, for example, a metal material such as aluminum die casting, and supports the first and second filter units **16** and **18**. In the case of the present embodiment, the casing **14** includes a protective glass **20** through which light from a subject passes.

[0030] As illustrated in FIG. 2, the imaging apparatus **10** includes an imaging element **22** including an imaging surface **22a**, on which light from a subject is incident, facing to the protective glass **20** in the extending direction of an optical axis LA of the imaging apparatus **10** (that is, the front-rear direction (X-axis direction) of the imaging apparatus **10**) at an interval. The imaging element **22** is a photoelectric conversion element such as a CCD or a CMOS, and creates image data of a subject on the basis of light from the subject incident on the imaging surface **22a** (an image of the subject) via the protective glass **20**. Note that the optical axis LA is orthogonal to the imaging surface **22a** of the imaging element **22** and passes through the center of the rectangular imaging surface **22a**.

[0031] In the case of the present embodiment, as illustrated in FIGS. 2 and 3, the first filter unit **16** includes a first optical filter **24** and a frame-shaped first frame structure **26** that supports an outer peripheral portion of the first optical filter **24**. In addition, the second filter unit **18** includes a second optical filter **28** and a second frame structure **30** that supports an outer peripheral portion of the second optical filter **28**.

[0032] In the case of the present embodiment, the first optical filter **24** is an electronic ND filter whose light transmittance can be changed, for example, a liquid crystal filter. The light transmittance of the first optical filter **24** changes by changing the driving voltage applied to the first optical filter **24**. The second optical filter **28** is a filter whose light transmittance cannot be changed, that is, light transmittance is fixed, for example, a glass. In the case of the present embodiment, the first and second optical filters **24** and **28** have a rectangular shape similarly to the imaging surface **22a** of the imaging element **22**.

[0033] The first filter unit **16**, that is, the first frame structure **26** supporting the first optical filter **24** is supported by the casing **14** so as to be rotatable about a first rotation center line C1 extending in the front-rear direction (X-axis direction) of the imaging apparatus **10**. The second filter unit **18**, that is, the second frame structure **30** supporting the second optical filter **28** is supported by the casing **14** so as to be rotatable about a second rotation center line C2 extending in the front-rear direction of the imaging apparatus **10**.

[0034] In the case of the present embodiment, as illustrated in FIGS. 2 and 3, the first and second rotation center lines C1 and C2 are located on the same straight line. Therefore, the first filter unit **16** rotates forward with respect to the second filter unit **18**. Since the first and second rotation center lines C1 and C2 are located on the same straight line, the support shaft **32** that rotatably supports the first and second filter units **16** and **18** can be shared.

[0035] As illustrated in FIG. 3, the filter module **12** includes a first drive mechanism **34** that rotates the first filter unit **16** about the first rotation center line C1, and a second drive mechanism **36** that rotates the second filter unit **18** about the second rotation center line C2.

[0036] In the case of the present embodiment, the first and second drive mechanisms **34** and **36** are so-called rack and pinion mechanisms.

[0037] The first drive mechanism **34** includes a first rack **38** that extends in the left-right direction (Y-axis direction) of the imaging apparatus **10** and is supported by the casing **14** so as to be movable in the left-right direction, and a first drive gear **40** that engages with the first rack **38** and moves the first rack **38** in the left-right direction. The first rack **38** engages with a pinion portion **26a** formed in the first frame structure **26** of the first filter unit **16**. The rotation of the first drive gear **40** moves the first rack **38** in the left-right direction, whereby the pinion portion **26a** rotates about the first rotation center line C1. As a result, the first filter unit **16** rotates about the first rotation center line C1.

[0038] The second drive mechanism **36** includes a second rack **42** that extends in the left-right direction (Y-axis direction) of the imaging apparatus **10** and is supported by the casing **14** so as to be movable in the left-right direction, and a second drive gear **44** that engages with the second rack **42** and moves the second rack **42** in the left-right direction. The second rack **42** extends behind and parallel to the first rack **38**. In addition, the second rack **42** engages with a pinion portion **30a** formed in the second frame structure **30** of the second filter unit **18**. The rotation of the second drive gear **44** moves the second rack **42** in the left-right direction, whereby a pinion portion **28a** rotates about the second rotation center line C2. As a result, the second filter unit **18** rotates about the second rotation center line C2.

[0039] By the first drive mechanism **34**, the first filter unit **16** rotates about the first rotation center line C1 between a first filtering position and a first retraction position. In addition, by the second drive mechanism **36**, the second filter unit **18** rotates about the second rotation center line C2 between a second filtering position and a second retraction position.

[0040] FIG. **4A** is a rear view of the filter module in a state where the first filter unit is located at the first filtering position and the second filter unit is located at the second retraction position. FIG. **4B** is a rear view of the filter module in a state where the first filter unit is located at the first retraction position and the second filter unit is located at the second filtering position. As illustrated in FIG. **4A**, the first filter unit **16** is rotated by the first drive mechanism **34** and disposed at the first filtering position. Specifically, when the first filter unit **16** is located at the first filtering position, the first optical filter **24** exists in front of the imaging surface **22a** of the imaging element **22**. As a result, light from the subject after passing through the protective glass **20** and before reaching the imaging surface **22a** passes through the first optical filter **24**. As a result, the light from the subject filtered by the first optical filter **24** is incident on the imaging surface **22a**.

[0041] As illustrated in FIG. **4B**, the first filter unit **16** is rotated by the first drive mechanism **34** and disposed at the first retraction position. Specifically, the first filter unit **16** retracts to a position deviated from the front of the imaging surface **22a** of the imaging element **22** as the first retraction position. In the case of the present embodiment, the first filter unit **16** retracts to the left (when viewed from the front of the imaging apparatus **10**) from the front of the imaging surface **22a**. As a result, light from the subject is incident on the imaging surface **22a** without being disturbed by the first filter unit **16**, that is, without passing through the first optical filter **24**.

[0042] As illustrated in FIG. **4B**, the second filter unit **18** is rotated by the second drive mechanism **36** and disposed at the second filtering position. Specifically, when the second filter unit **18** is located at the second filtering position, the second optical filter **28** exists in front of the imaging surface **22a** of the imaging element **22**. Light from the subject after passing through the protective glass **20** and before reaching the imaging surface **22a** passes through the second optical filter **28**. As a result, the light from the subject filtered by the second optical filter **28** is incident on the imaging surface **22a**. Note that the second filtering position is located behind the first filtering position.

[0043] As illustrated in FIG. **4A**, the second filter unit **18** is rotated by the second drive mechanism **36** and disposed at the second retraction position. Specifically, the second filter unit **18** retracts to a position deviated from the front of the imaging surface **22a** of the imaging element **22** as the second retraction position. In the case of the present embodiment, the second filter unit **18** retracts to the left (when viewed from the front of the imaging apparatus **10**) from the front of the imaging surface **22a**. As a result, light from the subject is incident on the imaging surface **22a** without being disturbed by the second filter unit **18**, that is, without passing through the second optical filter **28**. Note that the second retraction position is located behind the first retraction position.

[0044] In the case of the present embodiment, as illustrated in FIG. **4B**, the first filter unit **16** is rotated substantially by 90 degrees about the first rotation center line C1 by the first drive mechanism **34**. Therefore, the positional relationship between the first filtering position and the first retraction position is a positional relationship in which the first filter unit **16** located at one of

the first filtering position and the first retraction position rotates by 90 degrees to be disposed at the other. As a result, the posture (two-dot chain line) of the first optical filter **24** (**24'**) of the first filter unit **16** located at the first filtering position is different from the posture (solid line) of the first optical filter **24** located at the first retraction position by 90 degrees. That is, the longitudinal direction of the first optical filter **24** changes from the left-right direction (Y-axis direction) to the height direction (Z-axis direction) of the imaging apparatus **10**.

[0045] The first rotation center line **C1** of the first filter unit **16** is positioned such that the first optical filter **24** (**24'**) located at the first filtering position and the first optical filter **24** located at the first retraction position are adjacent to each other as much as possible without overlapping each other. For example, in the case of the present embodiment, the first rotation center line **C1** is positioned so as not to pass through the first optical filter **24** and near the lower left corner (when viewed from the front of the imaging apparatus **10**) of the first optical filter **24** (**24'**) when located at the first filtering position.

[0046] With such a first rotation center line **C1**, the movement range of the first optical filter **24**, that is, the first filter unit **16** can be made smaller than a case where the first optical filter **24** moves in parallel in the left-right direction (Y-axis direction) of the imaging apparatus **10**. As a result, it is possible to suppress an increase in size of the imaging apparatus **10**, particularly, an increase in size in the left-right direction. In addition, it is possible to suppress deterioration in designability of the imaging apparatus **10** due to an increase in size in the left-right direction.

[0047] Note that, in a case where the first optical filter **24** is moved in parallel in the height direction (Z-axis direction) of the imaging apparatus **10**, which is the lateral direction thereof, it is necessary to increase the size of the imaging apparatus **10** in the height direction, or it is necessary to reduce the size of the first optical filter **24**, that is, to reduce the size of the imaging element **22**. However, in that case, the designability or performance of the imaging apparatus **10** is greatly impaired.

[0048] The designability of the imaging apparatus **10** will be supplemented.

[0049] FIG. 5A is a front view schematically illustrating a layout of the imaging apparatus of the embodiment. FIG. 5B is a front view schematically illustrating a layout of the imaging apparatus of Comparative Example 1. Furthermore, FIG. 5C is a front view schematically illustrating a layout of the imaging apparatus of Comparative Example 2.

[0050] FIGS. 5A to 5C schematically illustrate layouts in a case where the first filter unit **16** rotates by 90 degrees (Example 1), a case where the first filter unit **16** moves in parallel in the left-right direction (Y-axis direction) of the imaging apparatus (Comparative Example 1), and a case where the first filter unit **16** moves in parallel in the height direction (Z-axis direction) (Comparative Example 2), respectively.

[0051] As illustrated in FIG. 5A, the imaging apparatus **10** is roughly divided into a main body portion **10a** and a grip portion **10b** when viewed from the front. A front silhouette (contour shape viewed from the front) of the main body portion **10a** greatly affects the designability of the imaging apparatus **10**. For example, in a case where the imaging apparatus **10** has a so-called single-lens reflex camera design, it is preferable that the size **r1** of the main body portion **10a** in the left-right direction (Y-axis direction) of the imaging apparatus **10** is about 1.2 to 1.8 times the size **r2** in the height direction (Z-axis direction). It is necessary to store the imaging element **22**, components related to the imaging element **22**, and the first filter unit **16** in the main body portion **10a** while maintaining the designability of the front silhouette of the main body portion **10a** of the imaging apparatus **10**. A battery (not illustrated) is stored in the grip portion **10b**.

[0052] As illustrated in FIG. 5A, the front silhouette of the main body portion **10a** of the imaging apparatus **10** has a rectangular shape in which the size **r1** of the imaging apparatus **10** in the left-right direction (Y-axis direction) is larger than the size **r2** in the height direction (Z-axis direction). Furthermore, in consideration of the designability of the imaging apparatus **10**, the optical axis **LA** of the imaging apparatus **10** passes through the center of the front silhouette of the rectangular main

body portion **10a**.

[0053] Furthermore, the main body portion **10a** of the imaging apparatus **10** needs to include a storage area SA for components related to the imaging element **22**. Examples of components related to the imaging element **22** include a substrate on which the imaging element **22** is mounted, a cooling device that cools the imaging element **22**, and the like. Furthermore, in a case where the imaging apparatus **10** has a camera shake correction function, an actuator that displaces the imaging element **22** in the left-right direction (Y-axis direction) and the height direction (Z-axis direction) of the imaging apparatus **10** is included in components related to the imaging element **22**. Therefore, the storage area SA has a predetermined positional relationship and a predetermined size with respect to the imaging element **22**.

[0054] Furthermore, a movement range MA (hatched portion) of the first filter unit **16** that rotates substantially by 90 degrees about the first rotation center line C1 needs to be included in the main body portion **10a** of the imaging apparatus **10**.

[0055] The sizes in the left-right direction (Y-axis direction) and the height direction (Z-axis direction) of the imaging apparatus **10** of the first filter unit **16** when the first filter unit **16** is located at the first filtering position are defined as s1 and s2, respectively. In this case, the sizes m1 and m2 in the left-right direction and the height direction of the movement range MA of the first filter unit **16** can be expressed by Mathematical Formulas 1 and 2.

$$\begin{array}{l} \text{[Math1]} \qquad \qquad \qquad \text{[Math2]} \\ \text{[00001]} \quad m1 = s1 + s2 \quad (\text{MathematicalFormula1}) \quad m2 = s1 + s2 \quad (\text{MathematicalFormula2}) \end{array}$$

[0056] The sizes r1 and r2 in the left-right direction (Y-axis direction) and the height direction (Z-axis direction) of the main body portion **10a** necessary to include the storage area SA and the movement range MA can be expressed by Mathematical Formulas 3 and 4.

$$\begin{array}{l} \text{[Math3]} \qquad \qquad \qquad \text{[Math4]} \\ \text{[00002]} \quad r1 = s1 + 2 \times s2 \quad (\text{MathematicalFormula3}) \quad r2 = 2 \times s1 + s2 \quad (\text{MathematicalFormula4}) \end{array}$$

[0057] In the case of the imaging apparatus **110** of Comparative Example 1 illustrated in FIG. 5B, the first filter unit **16** does not rotate and moves in parallel in the left-right direction (Y-axis direction) of the imaging apparatus **10**. When retracting from the front of the imaging element **22**, the first filter unit **16** moves in parallel to the left side of the main body portion **110a**. Because the battery exists in a grip portion **110b**, the first filter unit **16** cannot move into the grip portion **110b**.

[0058] In the case of the imaging apparatus **110** of Comparative Example 1 illustrated in FIG. 5B, the sizes m3 and m4 in the left-right direction (Y-axis direction) and the height direction (Z-axis direction) of the movement range MA of the first filter unit **16** can be expressed by Mathematical Formulas 5 and 6. In addition, the size r3 in the left-right direction of the main body portion **10a** necessary to include the storage area SA and the movement range MA can be expressed by Mathematical Formula 7.

$$\begin{array}{l} \text{[Math5]} \qquad \qquad \text{[Math6]} \qquad \qquad \text{[Math7]} \\ \text{[00003]} \quad m3 = 2 \times s1 \quad (\text{MathematicalFormula5}) \quad m4 = s2 \quad (\text{MathematicalFormula6}) \quad r3 = s1 \quad (\text{MathematicalFormula7}) \end{array}$$

[0059] Furthermore, in the case of the imaging apparatus **210** of Comparative Example 2 illustrated in FIG. 5C, the first filter unit **16** does not rotate and moves in parallel in the height direction (Z-axis direction) of the imaging apparatus **210**. When retracting from the front of the imaging element **22**, the first filter unit **16** moves in parallel to the upper side of the main body portion **110a**.

[0060] In the case of the imaging apparatus **210** of Comparative Example 2 illustrated in FIG. 5C, the sizes m5 and m6 in the left-right direction (Y-axis direction) and the height direction (Z-axis direction) of the movement range MA of the first filter unit **16** can be expressed by Mathematical Formulas 8 and 9. In addition, the size r4 in the height direction of the main body portion **210a** necessary to include the storage area SA and the movement range MA can be expressed by Mathematical Formula 10.

[00004]  $m_5 = s_1$  (MathematicalFormula10)

[0061] As illustrated in FIGS. 5A to 5C, when the first filter unit **16** rotates by 90 degrees about the first rotation center line **C1**, the area of the movement range **MA** necessary for the movement is larger than that in the case of parallel movement in the left-right direction (Y-axis direction) or the height direction (Z-axis direction).

[0062] However, the size  $m_1$  in the left-right direction of the movement range **MA** required for the first filter unit **16** in the embodiment to rotate by 90 degrees is smaller than the size  $m_3$  in the left-right direction of the movement range **MA** of the first filter unit **16** in Comparative Example 1, that is, smaller than twice the size  $s_1$  in the left-right direction of the first filter unit **16**. In addition, the size  $m_2$  in the height direction of the movement range **MA** of the first filter unit **16** in the embodiment is smaller than the size  $m_6$  in the height direction of the movement range **MA** of the first filter unit **16** in Comparative Example 2, that is, smaller than twice the size  $s_2$  in the height direction of the first filter unit **16**.

[0063] As a result, as illustrated in FIGS. 5A to 5C, in the case of the embodiment in which the first filter unit **16** rotates by 90 degrees about the first rotation center line **C1**, the influence on the designability of the imaging apparatus is small as compared with Comparative Example 1 and Comparative Example 2 in which the first filter unit **16** moves in parallel in the left-right direction (Y-axis direction) or the height direction (Z-axis direction). That is, the imaging apparatus can obtain a design of a so-called single-lens reflex camera. In the case of Comparative Example 1, the main body portion **110a** of the imaging apparatus **110** has a rectangular front silhouette extremely long in the left-right direction, and is separated from the design of the single-lens reflex camera. Furthermore, in the case of Comparative Example 2, the main body portion **210a** of the imaging apparatus **210** has a front silhouette close to a square, and is separated from the design of the single-lens reflex camera.

[0064] In the case of the present embodiment, as illustrated in FIG. 4B, the second filter unit **18** is rotated substantially by 90 degrees about the second rotation center line **C2** by the second drive mechanism **36**. Therefore, the positional relationship between the second filtering position and the second retraction position is a positional relationship in which the second filter unit **18** located at one of the second filtering position and the second retraction position rotates by 90 degrees to be disposed at the other. As a result, the posture (solid line) of the second optical filter **28** of the second filter unit **18** located at the second filtering position is different from the posture (two-dot chain line) of the second optical filter **28** (**28'**) located at the second retraction position by 90 degrees. That is, the longitudinal direction of the second optical filter **28** changes from the left-right direction (Y-axis direction) to the height direction (Z-axis direction) of the imaging apparatus **10**.

[0065] The second rotation center line **C2** of the second filter unit **18** is positioned such that the second optical filter **28** located at the second filtering position and the second optical filter **28** (**28'**) located at the second retraction position are adjacent to each other as much as possible without overlapping each other. For example, in the case of the present embodiment, the second rotation center line **C2** is positioned so as not to pass through the second optical filter **28** and near the lower left corner (when viewed from the front of the imaging apparatus **10**) of the second optical filter **28** when located at the second filtering position.

[0066] With such a second rotation center line **C2**, the movement range of the second optical filter **28**, that is, the second filter unit **18** can be made smaller than a case where the second optical filter **28** moves in parallel in the left-right direction (Y-axis direction) of the imaging apparatus **10**. As a result, it is possible to suppress an increase in size of the imaging apparatus **10**, particularly, an increase in size in the left-right direction. In addition, it is possible to suppress deterioration in designability of the imaging apparatus **10** due to an increase in size in the left-right direction.

[0067] Note that, in a case where the second optical filter **28** is moved in parallel in the height



direction (Z-axis direction) of the imaging apparatus **10**, which is the lateral direction thereof, it is necessary to increase the size of the imaging apparatus **10** in the height direction, or it is necessary to reduce the size of the second optical filter **28**, that is, to reduce the size of the imaging element **22**. However, in that case, the designability or performance of the imaging apparatus **10** is greatly impaired.

[0068] Similarly to the first filter unit **16**, the second filter unit **18** also rotates substantially by 90 degrees, and thus does not greatly affect the designability of the imaging apparatus **10** as compared with the case of parallel movement in the left-right direction (Y-axis direction) or the height direction (Z-axis direction) of the imaging apparatus **10**.

[0069] That is, the size in the left-right direction of the movement range of the second filter unit **18** required to rotate by 90 degrees about the second rotation center line C2 is smaller than twice the size in the left-right direction of the second filter unit **18** when the second filter unit **18** is located at the second filtering position. The size in the height direction of the movement range of the second filter unit is smaller than twice the size in the height direction of the second filter unit **18** when the second filter unit **18** is located at the second filtering position. As a result, the imaging apparatus **10** can obtain a design of a so-called single-lens reflex camera.

[0070] In the case of the present embodiment, the rotation operation of the first filter unit **16** by the first drive mechanism **34** and the rotation operation of the second filter unit **18** by the second drive mechanism **36** are synchronized.

[0071] Specifically, as illustrated in FIG. 4A, when the first filter unit **16** is located at the first filtering position (that is, the position in front of the imaging surface **22a** of the imaging element **22**), the rotation operations of the first and second filter units **16** and **18** are synchronized such that the second filter unit **18** is located at the second retraction position (that is, a position deviated from the front of the imaging surface **22a**). In addition, as illustrated in FIG. 4B, when the first filter unit **16** is located at the first retraction position, the rotation operations of the first and second filter units **16** and **18** are synchronized such that the second filter unit **18** is located at the second filtering position. Therefore, the first drive mechanism **34** and the second drive mechanism **36** are synchronized.

[0072] In the case of the present embodiment, as illustrated in FIGS. 2 and 3, the first drive gear **40** of the first drive mechanism **34** and the second drive gear **44** of the second drive mechanism **36** engage with each other. The second drive gear **44** is engaged with a power transmission gear **46** which is a power source for supplying power to the first drive mechanism **34** and the second drive mechanism **36**. The power transmission gear **46** is connected to a rotary knob **48** disposed on the front surface of the imaging apparatus **10** and rotated by a user.

[0073] When the user rotates the rotary knob **48** from the state illustrated in FIG. 4A, the power transmission gear **46** rotates, so that the first drive gear **40** and the second drive gear **44** rotate synchronously. As a result, the first rack **38** engaging with the first drive gear **40** moves in the right direction (when viewed from the front of the imaging apparatus **10**), and the second rack **42** engaging with the second drive gear **44** moves in the opposite direction, that is, in the left direction. As the first rack **38** and the second rack **42** move in opposite directions to each other, the first filter unit **16** and the second filter unit **18** rotate in opposite directions to each other.

[0074] FIGS. 6A to 6D are rear views of the filter module in a state where the second filter unit rotates toward the second filtering position while the first filter unit rotates toward the first retraction position.

[0075] As illustrated in FIGS. 6A to 6D, when the user rotates the rotary knob **48** forward from the state illustrated in FIG. 4A, the first filter unit **16** rotates from the first filtering position toward the first retraction position, and the second filter unit **18** rotates from the second retraction position toward the second filtering position. During the rotation, the first filter unit **16** and the second filter unit **18** overlap in the front-rear direction (X-axis direction) of the imaging apparatus **10**. In the present embodiment, the second filter unit **18** passes behind the first filter unit **16**.

[0076] When the user reversely rotates the rotary knob **48** from the state illustrated in FIG. **4B**, the second filter unit **18** rotates from the second filtering position toward the second retraction position while the first filter unit **16** rotates from the first retraction position toward the first filtering position.

[0077] By such synchronous operation of the first drive mechanism **34** and the second drive mechanism **36**, the first filter unit **16** and the second filter unit **18** can be simultaneously rotated only by rotating one rotary knob **48**. That is, the user can dispose the first optical filter **24** or the second optical filter **28** in front of the imaging surface **22a** of the imaging element **22** only by rotating the rotary knob **48** in the forward direction or the reverse direction. As a result, the user can easily change the optical filter to be used from the first optical filter **24** to the second optical filter **28** or vice versa.

[0078] Note that, in the case of the present embodiment, the first optical filter **24** is an electronic ND filter having a changeable light transmittance, and the second optical filter **28** is a filter having a fixed light transmittance, for example, glass. The second optical filter **28** has an optical path length substantially identical to the optical path length of the first optical filter **24** so that the focus does not change when the first optical filter **24** is changed to the second optical filter **28**.

[0079] Furthermore, in the case of the present embodiment, the imaging apparatus **10** includes a sensor unit **50** that detects the light transmittance of the first optical filter **24**.

[0080] FIG. **7** is an exploded rear perspective view of the first filter unit. FIG. **8** is a front perspective view of a ND filter assembly. FIG. **9** is a partial cross-sectional view of the ND filter assembly.

[0081] As illustrated in FIGS. **7** to **9**, the sensor unit **50** is provided in the first filter unit **16**. The sensor unit **50** includes a light source **52** such as an LED, a light guide member **54** that disperses light output from the light source **52** and emits first and second light **L1** and **L2**, a first optical sensor **56** that detects the intensity of the first light **L1**, and a second optical sensor **58** that detects the intensity of the second light **L2**.

[0082] In the case of the present embodiment, in the first filter unit **16**, the first optical filters **24** are provided between glass plates **60** and **62** and between glass plates **64** and **66**, respectively. The one first light **L1** is transmitted through the two first optical filters **24** and the four glass plates **60** to **66** and is incident on the first optical sensor **56**. The other second light **L2** is transmitted only through the glass plates **62** and **64** and is incident on the second optical sensor **58**.

[0083] When the light transmittance of the first optical filter **24** is a predetermined light transmittance (that is, when the driving voltage applied to the first optical filter **24** is a predetermined voltage), the intensity difference between the intensity of the first light **L1** and the intensity of the second light **L2** detected by the first optical sensor **56** is a predetermined intensity difference. When the driving voltage of the first optical filter **24** is a predetermined voltage but the intensity difference between the first light **L1** and the second light **L2** is different from the predetermined intensity difference, it can be determined that an abnormality such as deterioration or failure has occurred in the first optical filter **24**.

[0084] In the case of the present embodiment, the sensor unit **50** is provided in the first filter unit **16**. Therefore, the sensor unit **50** can detect the light transmittance of the first optical filter **24** regardless of the position of the first filter unit **16**.

[0085] According to the present embodiment as described above, optical filters having various light transmittances can be used in the imaging apparatus without increasing the size of the imaging apparatus.

[0086] Specifically, unlike the present embodiment, when a plurality of optical filters having different light transmittances are used, a retraction space for a remaining plurality of optical filters that are not used is required. On the other hand, according to the present embodiment, since the first optical filter **24** is one electronic filter capable of changing the light transmittance, the retraction space can be reduced.

[0087] Furthermore, in the case of the present embodiment, the first optical filter **24** has a rectangular shape and rotates by 90 degrees toward the first retraction position. As a result, it is possible to suppress an increase in size of the imaging apparatus **10** in the left-right direction (Y-axis direction) as compared with a case where the first optical filter **24** moves in parallel in the longitudinal direction (Y-axis direction) and retracts.

[0088] Furthermore, in the case of the present embodiment, as illustrated in FIGS. **4A** and **4B**, the second retraction position of the second optical filter **28** (second filter unit **18**) is located behind the first retraction position of the first optical filter **24** (first filter unit **16**), that is, arranged in the front-rear direction (X-axis direction) of the imaging apparatus **10**. As a result, as compared with a case where the first retraction position and the second retraction position are arranged in the left-right direction (Y-axis direction) or the height direction (Z-axis direction) of the imaging apparatus **10**, it is possible to suppress deterioration in designability of the imaging apparatus **10**, that is, deterioration in designability of a so-called single-lens reflex camera.

[0089] Although the embodiment of the present disclosure has been described above with reference to the above-described embodiment, the embodiment of the present disclosure is not limited to the above-described embodiment.

[0090] For example, in the case of the above-described embodiment, as illustrated in FIG. **3**, the imaging apparatus **10** includes the second optical filter **28** that is glass or the like whose light transmittance cannot be changed. The second optical filter **28** has an optical path length substantially identical to the optical path length of the first optical filter **24**. Accordingly, even when the first optical filter **24** is changed to the second optical filter **28**, it is possible to maintain a focused state. However, the embodiment of the present disclosure is not limited thereto. If a focused state can be maintained by another mechanism even if the first optical filter **24** is retracted from the first filtering position (position in front of the imaging surface **22a** of the imaging element **22**), the second optical filter **28**, that is, the second filter unit **18** can be omitted.

[0091] In the case of the above-described embodiment, as illustrated in FIG. **3**, the first and second drive mechanisms **34** and **36** that rotate the first and second filter units **16** and **18** are so-called rack and pinion mechanisms. However, the embodiment of the present disclosure is not limited thereto. For example, two motors may rotate the first and second filter units **16** and **18**, respectively.

[0092] Furthermore, in the case of the above-described embodiment, the first and second filter units **16** and **18** are rotated by the user rotating the rotary knob **48**, that is, manually rotated. However, the embodiment of the present disclosure is not limited thereto. For example, when the user presses the button, the motor may rotate the first and second filter units **16** and **18**. Furthermore, in the case of the above-described embodiment, as illustrated in FIGS. **7** to **9**, the sensor unit **50** that detects the light transmittance of the first optical filter **24** is provided in the first filter unit **16**. However, the embodiment of the present disclosure is not limited thereto. The sensor unit **50** may be provided in the casing **14** of the filter module **12**. In this case, the sensor unit **50** is provided at a position on the casing **14** where the light transmittance of the first optical filter **24** can be detected when the first filter unit **16** is located at the first filtering position (FIG. **4A**).

[0093] That is, in a broad sense, an imaging apparatus according to an embodiment of the present disclosure includes: an imaging element including an imaging surface on which light from a subject is incident; a first filter unit including an electronic first optical filter capable of changing a light transmittance; and a first drive mechanism that rotates the first filter unit about a first rotation center line between a first filtering position and a first retraction position, in which the first filtering position is a position where the first optical filter exists in front of the imaging surface of the imaging element and the light before reaching the imaging surface passes through the first optical filter, and in which the first retracted position is a position where the first filter unit is out of the front of the imaging surface.

[0094] As described above, the above-described embodiment has been described as an example of the technique in the present disclosure. To that end, the drawings and the detailed description are

provided. Therefore, the components described in the drawings and the detailed description may include not only components essential for solving the problem but also components that are not essential for solving the problem in order to describe the above-described technology. Therefore, it should not be immediately recognized that these non-essential components are essential based on the fact that these non-essential components are described in the drawings and the detailed description.

[0095] Further, the above-described embodiment is provided to describe the technique in the present disclosure, and hence it is possible to make various changes, replacements, additions, omissions, and the like within the scope of claims or the equivalent thereof.

[0096] The present disclosure is applicable to an imaging apparatus including a plurality of ND filters having different light transmittances.

## Claims

1. A filter module usable in conjunction with an imaging apparatus including an imaging element, the filter module comprising: a first optical filter; a first support member supporting the first optical filter; and a first drive unit moving the first support member, wherein the filter module is configured to be positioned along an optical axis of the imaging element included in the imaging apparatus, the first optical filter being rotationally movable by the first drive unit between a first position and a second position, wherein the first position is a position within an imaging range of the imaging element, and wherein the second position is a position out of the imaging range of the imaging element.
2. The filter module according to claim 1, wherein the first drive unit includes a first drive mechanism and a first power source supplying power to the first drive mechanism.
3. The filter module according to claim 2, wherein the first power source is a motor.
4. The filter module according to claim 2, wherein the first power source is a rotary knob operated by a user.
5. The filter module according to claim 1, further comprising: a second optical filter; a second support member supporting the second optical filter; and a second drive unit moving the second support member, wherein the second optical filter being rotationally movable by the second drive unit between the first position and the second position.
6. The filter module according to claim 5, wherein when the first optical filter is located at the first position, the second optical filter is located at the second position, and wherein when the second optical filter is located at the first position, the first optical filter is located at the second position.
7. The filter module according to claim 6, wherein the second drive unit includes a second drive mechanism and a second power source supplying power to the second drive mechanism.
8. The filter module according to claim 5, wherein the first drive unit includes a first drive mechanism and a first power source supplying power to the first drive mechanism, wherein the second drive unit includes a second drive mechanism and a second power source supplying power to the second drive mechanism, and wherein the first and second power sources are a common power source.
9. The filter module according to claim 5, wherein the first and second optical filters is rotated about a common rotation center line.
10. The filter module according to claim 1, wherein the first optical filter is rotatable by 90 degrees between the first and second positions.
11. The filter module according to claim 5, wherein the second optical filter is rotatable by 90 degrees between the first and second positions.
12. The filter module according to claim 9, wherein as viewed in an extending direction of an optical axis of the imaging apparatus, the rotation center line is closer to a vertex of a rectangular-shaped imaging range than the optical axis.

**13.** The filter module according to claim 1, wherein when the imaging apparatus including a lens mount to which a lens device is detachably attached, the first optical filter is located in an extending direction of an optical axis of the imaging apparatus between imaging element and the lens mount.

**14.** The filter module according to claim 5, wherein when the imaging apparatus including a lens mount to which a lens device is detachably attached, the second optical filter is located in an extending direction of an optical axis of the imaging apparatus between imaging element and the lens mount.

**15.** The filter module according to claim 5, wherein the first optical filter is an optical filter having a changeable light transmittance, and wherein the second optical filter is an optical filter having a fixed light transmittance.

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